

National Forensic Sciences University

School of Cybersecurity and Digital Forensics

M.Sc. DFSI - Semester I

MSDF26-PC-103 Programming for Digital Forensics

Lab Assignment #2: Linux Commands for Digital Forensic Investigation

Objective

1. Understand the role of Linux commands in digital forensic investigations.
2. Navigate the Linux filesystem and identify files and directories relevant to an investigation.
3. Create, copy, move, rename, and examine files without modifying the original evidence.
4. View file metadata, permissions, ownership, size, and timestamps.
5. Search for files and extract relevant information using Linux commands.
6. Calculate and verify cryptographic hash values to maintain evidence integrity.
7. Analyse text files and system logs using command-line utilities.
8. Redirect command output to files and prepare a basic forensic investigation report.

Sr No	Description
Task-1	Create the Forensic Workspace
	Create the following directory structure using Linux commands: <pre>DF_Lab_Assignment_2/ ├── evidence/ ├── working_copy/ ├── extracted/ ├── logs/ └── report/</pre> Perform the following: 1. Display the current working directory. 2. Create the required directories. 3. Display the directory structure. 4. Navigate between the directories using relative and absolute paths. 5. Save the directory listing in directory_structure.txt.

	Suggested commands: pwd, mkdir, cd, ls
Task-2	Evidence File Creation and Management
	Create the following sample files inside the evidence directory: • system_log.txt • login_records.txt • user_notes.txt • suspicious_script.sh • .hidden_note.txt Perform the following: 1. Create the files using appropriate Linux commands. 2. Add sample content to each file. 3. Display all files, including hidden files. 4. Copy the evidence files to the working_copy directory. 5. Rename the copied user_notes.txt file to examined_notes.txt. 6. Compare the contents of the original and copied files. Suggested commands: touch, echo, ls -la, cp, mv, diff
Task-3	Examine File Metadata
	Examine the original evidence files and record the following information: • Filename • File type • File size • Owner and group • File permissions • Last access time • Last modification time • Metadata change time Save the metadata output in logs/file_metadata.txt. Suggested commands: ls -lh, ls -l, stat, file
Task-4	File Permissions and Ownership Analysis
	Examine the permissions assigned to suspicious_script.sh. Perform the following: 1. Display its current permissions. 2. Identify permissions assigned to the owner, group, and others. 3. Add execute permission for the file owner. 4. Remove write permission from the group and others. 5. Display the updated permissions in symbolic and numeric forms. 6. Explain the forensic importance of file permissions and ownership. Suggested commands: ls -l, chmod, stat Perform permission changes only on the file stored in the working_copy directory.

Task-5	Search for Files and Directories
	Search the forensic workspace for: 1. All .txt files. 2. All .sh files. 3. Hidden files. 4. Files containing the word password. 5. Files modified during the last 24 hours. 6. Files larger than a specified size. 7. Files having executable permission. Save the search results in logs/file_search_results.txt. Suggested commands: find, grep, ls
Task-6	Analyse Text and Log Files
	Use Linux text-processing commands to examine system_log.txt and login_records.txt. Perform the following: 1. Display the complete contents of each file. 2. Display only the first five lines. 3. Display only the last five lines. 4. Count the number of lines, words, and characters. 5. Search for the following keywords: - o failed - o error - o unauthorized - o login - o root 6. Perform a case-insensitive keyword search. 7. Display matching lines along with their line numbers. 8. Save all matching entries in extracted/suspicious_entries.txt. Suggested commands: cat, less, head, tail, wc, grep
Task-7	Sort and Extract Investigation Data
	Using login_records.txt, perform the following: 1. Sort all records alphabetically. 2. Display only unique records. 3. Count repeated login entries. 4. Extract selected fields from each record. 5. Arrange the records based on a selected field. 6. Save the processed results in extracted/processed_login_records.txt. Suggested commands: sort, uniq, cut
Task-8	Hash Calculation and Integrity Verification

	Calculate cryptographic hash values for every file in the evidence directory. Perform the following: 1. Calculate the MD5 hash of each evidence file. 2. Calculate the SHA-256 hash of each evidence file. 3. Save the original hash values in logs/original_hashes.txt. 4. Copy the files to the working_copy directory. 5. Calculate the SHA-256 hashes of the copied files. 6. Compare the original and copied hash values. 7. Modify one copied file and calculate its hash again. 8. Explain why its hash value changed. Suggested commands: md5sum, sha256sum, cp, diff
Task-9	Command History and Audit Trail
	Maintain a record of the commands executed during the investigation. Perform the following: 1. Display the terminal command history. 2. Save the command history in logs/command_history.txt. 3. Record the current username, date, time, hostname, and working directory. 4. Save this information in logs/investigation_details.txt. 5. Explain the importance of maintaining an investigation audit trail. Suggested commands: history, whoami, date, hostname, pwd
Task-10	Archive the Investigation Files
	After completing the investigation: 1. Create a compressed archive of the logs, extracted, and report directories. 2. Name the archive forensic_investigation_<student_roll_number>.tar.gz. 3. Display the contents of the archive without extracting it. 4. Calculate the SHA-256 hash of the archive. 5. Save the archive hash in archive_hash.txt. Suggested commands: tar, sha256sum, ls -lh

Submission

Prepare a report containing the following sections:

1. Student name and enrolment number
2. Date and time of investigation
3. Objective of the investigation
4. Description of the evidence files
5. Commands used
6. File metadata observations
7. Permission analysis
8. Search results

9. Suspicious log entries
10. Original and working-copy hash values
11. Integrity verification result
12. Important findings
13. Conclusion

Marking Rubric:

Total Marks: 50, Each task carries 5 marks.

Assessment criterion	Marks
Creation and organisation of forensic workspace	4
Evidence file handling and creation of working copies	5
Examination of file type, metadata and timestamps	5
File permission and ownership analysis	4
File and keyword searching	5
Text and log-file analysis	6
Sorting and extraction of relevant records	4
Hash calculation and integrity verification	7
Command history and investigation audit trail	4
Archive creation and integrity verification	3
Final report, observations and screenshots	3
Total	50